Wumpus No Longer Alone

Problem Statement

1. Background

The classic Wumpus World pits a lone explorer against a single lurking beast. That era is over.

You have stumbled into an ancient cave system—one that has grown restless. Multiple Wumpuses now stalk its corridors in synchronized, oscillating patterns. Pits yawn in the darkness. Strange temporal anomalies warp your sense of time. The gold still gleams somewhere ahead, but the question is no longer "Can I survive?"—it is "Is survival even guaranteed before I take a single step?"

2. Problem Statement

You are given an $n \times n$ cave grid. An agent begins at cell $(1, 1)$ and must reach the cell containing the Gold (`G`). Before the agent moves, you must determine whether a guaranteed-safe path to the gold exists, accounting for Wumpus movement over time, pits, time zones, and stench accumulation.

If such a path exists, output `SAFE` followed by:

(a) The optimal path that minimizes total time cost.

If no such path exists, output `UNSAFE`.

3. Grid Specification

- The grid is 1-indexed: cell $(1, 1)$ is the top-left; cell (n, n) is the bottom-right.
- $5 \leq n \leq 15$.
- The agent moves orthogonally (up, down, left, right), one cell per turn.
- $(1, 1)$ and the cell containing `G` are always guaranteed safe—no Wumpus, Pit, or Time Zone will ever occupy these cells.

Each cell contains exactly one of the following symbols:

Symbol	Name	Description
`.`	Empty	Safe to enter; no effect.
`W`	Wumpus	Deadly. Entering a cell occupied by a Wumpus kills the agent immediately.
`P`	Pit	Entering incurs a +5 second time penalty.
`T`	Time Zone	Entering reduces the agent's accumulated time by 3 seconds (a benefit).
`G`	Gold	The goal cell. Reaching it ends the journey successfully.

4. Perception Rules

The agent perceives adjacent hazards without entering the cell. Percepts appear in the four orthogonally adjacent cells of a hazard.

4.1. Stench

A Wumpus at cell (i, j) emits Stench into all of its orthogonal neighbours:

$$(i-1, j), \quad (i+1, j), \quad (i, j-1), \quad (i, j+1)$$

The Wumpus's own cell does not carry a stench marker—only its neighbours do.

4.2. Breeze (Perceived)

A Pit at cell (i, j) emits a perceived Breeze into all of its orthogonal neighbours:

$$(i-1, j), \quad (i+1, j), \quad (i, j-1), \quad (i, j+1)$$

Percepts are additive: a cell adjacent to both a Wumpus and a Pit carries both Stench and perceived Breeze simultaneously.

4.3. Stench Accumulation—Death Rule

If the agent passes through 3 or more stench-bearing cells over the course of a path (cells that are adjacent to a Wumpus at the time of entry), the agent dies.

Important: Stench is dynamic and evaluated at the exact turn the agent enters a cell.

5. Wumpus Movement

Wumpuses move horizontally only, following a strict oscillating pattern determined by their fixed row index i. Movement is bounded by the grid perimeter: if a Wumpus would move beyond the boundary, it stays at the boundary cell for the remainder of that step.

5.1. Turn Structure

Movement alternates between Turn A and Turn B, cycling indefinitely.

When i is even	
Phase	**Steps**
Turn A	Move right 3, then left 1
Turn B	Move right 1, then left 3
When i is odd	
Phase	**Steps**
Turn A	Move left 3, then right 1
Turn B	Move left 1, then right 3

Let the net displacement per full cycle (Turn A + Turn B) be:

$$\Delta_{\text{even}} = (+3 - 1) + (+1 - 3) = 0,$$
$$\Delta_{\text{odd}} = (-3 + 1) + (-1 + 3) = 0.$$

The oscillation is zero-net: over any complete cycle the Wumpus returns to its original column (modulo boundary clamping).

- **Step-by-step resolution:** Each individual step within a turn is resolved sequentially. If the Wumpus hits a wall mid-sequence, it halts at the boundary and resumes from that position for subsequent steps in the same turn.
- **Row tracking:** Wumpuses move strictly horizontally and never change rows throughout the simulation.
- **Turn order:** All Wumpus positions are updated before the agent executes their move each turn.

6. Time Cost Model

The agent's journey accumulates time T according to the following rules:

Event	Time Effect
Moving into any cell (base cost)	+1 second
Entering a Pit `P`	+5 seconds
Entering a cell with Breeze	+2 seconds
Entering a Time Zone `T`	−3 seconds
Entering a Wumpus cell `W`	Agent dies
Stench count ≥ 3	Agent dies

The total change δ_{cell} for entering a cell is the base cost (+1) plus all applicable cell effects and penalties (e.g., entering a Pit with breeze costs $1 + 5 + 2 = +8$). After each move, accumulated time is updated as:

$$T \leftarrow \max(0,\ T + \delta_{\text{cell}})$$

Time is clamped to zero and cannot become negative.

7. Constraints

Parameter	Bound
Grid size n	$5 \leq n \leq 15$
Number of Wumpuses	$0 \leq \lvert W \rvert \leq n$
Number of Pits	$0 \leq \lvert P \rvert \leq \lfloor n^2/5 \rfloor$
Number of Time Zones	$0 \leq \lvert T \rvert \leq \lfloor n^2/5 \rfloor$
Turn limit	$4 \times n^2$

A path exceeding $4n^2$ turns is considered **invalid** (the agent is presumed lost).

8. Input Format

```
n
grid row 1
grid row 2
...
grid row n
```

- The grid is given as n space-separated rows, each containing n symbols.
- Exactly one cell contains `G`.
- Cell $(1, 1)$ is always `.` (the agent's starting position).
- There may be zero or more Wumpuses.

Example Input

```
5
. . . P .
W . . . .
. P . . G
. . W . .
. . . T .
```

9. Output Format

Line 1: `SAFE` or `UNSAFE`

If `UNSAFE`, print only `UNSAFE` and terminate.

If `SAFE`, print:

```
Line 1: SAFE
Line 2: <minimum time cost>
Line 3: <optimal path as sequence of (row,col) pairs>
```

The path on Line 3 must be printed as space-separated coordinate pairs:

```
(1,1) (1,2) (1,3) (1,4) (2,4) (3,4) (3,5)
```

If multiple paths share the minimum cost, print the lexicographically smallest one (compare coordinate pairs left to right).

Example Output

```
SAFE
3
(1,1) (1,2) (1,3) (1,4) (1,5) (2,5) (3,5)
```

10. Notes & Clarifications

1. A path through a Pit is valid—it incurs a penalty but does not kill the agent.
2. The stench death rule counts unique stench-bearing cells entered, not the number of Wumpuses nearby. A cell adjacent to two Wumpuses simultaneously counts as one stench encounter.
3. Two paths are distinct if they differ in at least one cell visited or in the order of visiting.
4. The agent cannot revisit a cell within a single path (no cycles).
5. The Gold cell `G` is always safe and incurs no cost beyond the base movement cost of +1.